

Aim:

To design and implement a radar signal detection system using matched filtering in MATLAB.

Apparatus / Software Requirements:

Computer/Laptop with MATLAB installed.

Theory:

RADAR (Radio Detection and Ranging) is a system that uses electromagnetic waves to detect and locate objects like aircraft, ships, and vehicles. It works by sending out signals and receiving the reflected echoes from targets.

- Optimal Detection: Best linear filter to maximize SNR in noisy conditions (AWGN).
- Pulse Compression: Uses long pulses for high energy while keeping short-pulse resolution.
- Processing Gain: Improves signal strength by a factor of E/N_0 (signal energy to noise ratio).

It provides:

- Range (distance): From time delay of the echo
- Velocity: From Doppler shift
- Position (angle): From antenna direction

The Radar Range Equation:

The performance of any radar system is governed by the Radar Range Equation, which relates the range of the target to the characteristics of the transmitter, receiver, antenna, and the target's reflectivity (Radar Cross Section). The power received (P_r) is expressed as:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

Where:

- P_t : Transmitted power
- G : Antenna gain
- λ : Wavelength
- σ : Radar Cross Section (RCS) of the target
- R : Range to the target

Key Challenges in Detection:

The main problem in radar systems is **noise**. Every radar receiver has thermal noise, which can hide weak signals coming from far or small targets. To detect a target properly, the radar needs a **high Signal-to-Noise Ratio (SNR)**. Earlier, this was improved by increasing the transmitted power. But this method has limits due to hardware restrictions and atmospheric effects.

Radar Detection Principle:

The Core Detection Cycle

- **Transmission:** Radar sends a high-power electromagnetic pulse using an antenna.
- **Propagation:** The pulse travels through air at the speed of light (3×10^8 m/s).
- **Reflection (Echo):** When the pulse hits an object, part of it reflects back as an echo.
- **Reception:** The radar receives the weak echo, amplifies it, and processes it to find the target's distance and properties.

Extracting Target Information (Simplified):

Radar calculates key details from the returned signal:

A. Range (Distance)

- Measured using time delay (Time of Flight)
- Signal travels to target and back

B. Angular Position

- Determined by the direction of the antenna
- Target lies in the beam where the echo is received

C. Velocity (Doppler Effect)

- Movement causes a change in frequency
 - Approaching target: Frequency increases
 - Moving away: Frequency decreases

The Detection Threshold

- If the received signal strength exceeds the threshold, the system declares a "**Hit**" or "Target Present."
- If the threshold is too low, the radar suffers from **False Alarms** (detecting noise as a target).
- If the threshold is too high, it might miss real targets (**Missed Detection**).

Factors Affecting Detection:

- Transmitted Power: Higher power allows for detection at longer ranges.
- Radar Cross Section (RCS): A measure of how "visible" a target is to radar. A large tanker has a high RCS, while a stealth fighter has a very low RCS.
- Antenna Gain: How effectively the antenna concentrates energy into a narrow beam.

Pulse Compression and Matched Filtering:

1. Range vs Resolution Trade-off

Long pulses provide high energy for long-range detection but give poor resolution. Short pulses give high resolution but have low energy, limiting detection range.

2. Pulse Compression Technique

A long coded pulse (frequency/phase modulated) is transmitted to maintain high energy. After processing, it behaves like a short pulse, achieving both high resolution and long range.

3. Matched Filtering

The matched filter compares the received signal with the transmitted reference signal. It produces a sharp peak for matches and reduces noise, maximizing SNR.

4. Signal Model

The received signal is given by $x(t) = s(t) + n(t)$.

The matched filter helps recover the original signal $s(t)$ from noisy input $x(t)$.

Mathematical Derivation of Matched Filter:

1. Output of the Filter

The output signal is obtained by convolving the input signal with the filter response.

$$y(t) = x(t) * h(t)$$

2. Objective

The goal is to design a filter that maximizes the Signal-to-Noise Ratio (SNR) at a specific time T .

This ensures the signal is strongest and noise is minimized at the sampling instant.

3. Impulse Response of Matched Filter

The optimal filter is a time-reversed and delayed version of the transmitted signal.

$$h(t) = k \cdot s^*(T - t)$$

Where:

k = constant (usually 1)

T = duration of the signal
 $s^*(t)$ = complex conjugate of the signal

4. Correlation Property

Matched filtering works like cross-correlation between the received signal and the transmitted signal.

When both signals align, a strong peak is produced, while noise remains low.

5. Frequency Domain Interpretation

In the frequency domain, the filter is the complex conjugate of the signal spectrum.

$$H(f) = S^*(f)e^{-j2\pi fT}$$

This means the Matched Filter:

1. Matches the Phase: It acts as a "phase conjugate" filter, aligning all spectral components of the signal to be in phase at time T .
2. Weights the Amplitude: It gives more weight to frequencies where the signal has high energy and less weight to frequencies where the signal is weak (and noise dominates).

The Matched Filter Algorithm:

The algorithm follows a "Time-Domain Correlation" approach. Here are the logical steps:

1. Define Parameters: Initialize sampling frequency (f_s), pulse width (T), and chirp bandwidth (B).
2. Reference Generation: Create a local copy (replica) of the transmitted LFM chirp signal.
3. Signal Acquisition: Capture the incoming raw data (which contains the delayed echo plus environmental noise).
4. Filter Construction: Create the impulse response $h(n)$ by time-reversing the complex conjugate of the reference signal.
5. Convolution/Correlation: Convolve the received signal with the impulse response.
6. Envelope Detection: Take the absolute value (magnitude) of the result to eliminate phase oscillations.
7. Thresholding (Optional): Compare the output against a threshold to declare a "Detection."

CODE:

% 1. Basic Setup

```
fs = 1e6;           % Sampling Frequency (1 MHz)

T = 100e-6;         % Pulse Duration (100 microseconds)

B = 2e5;            % Bandwidth (200 kHz)

c = 3e8;            % Speed of light (m/s)

% Generate the Radar Pulse (LFM Chirp)
```

```

t_pulse = 0:1/fs:T-1/fs;
s = chirp(t_pulse, 0, T, B);

%% 2. Create the Hidden Target

target_range = 45000;          % Target is 45km away

delay = (2 * target_range) / c; % Time for signal to go and come back

delay_samples = round(delay * fs);

% Create a 1ms timeline and hide the pulse in it

t_total = 0:1/fs:1e-3;

rx_signal = zeros(size(t_total));

    rx_signal(delay_samples : delay_samples + length(s) - 1) = s;

% Add random noise

noisy_rx = rx_signal + 1.2 * randn(size(t_total));

%% 3. Apply Matched Filter (The Detection Part)

% Flip the pulse to create the filter

h = fliplr(s);

% Correlate the noisy signal with our pulse template

% IMPORTANT: If you get an error here, rename your "conv.m" file!

mf_output = conv(noisy_rx, h, 'same');

mf_magnitude = abs(mf_output);

%% 4. Plot Results

figure;

% Plot the noisy signal (The pulse is invisible here)

subplot(2,1,1);

plot(t_total*1e3, noisy_rx);

title('Step 1: Noisy Received Signal (Target is hidden)');

xlabel('Time (ms)'); ylabel('Amplitude');

```

```
% Plot the filtered signal (The pulse becomes a clear peak)

subplot(2,1,2);

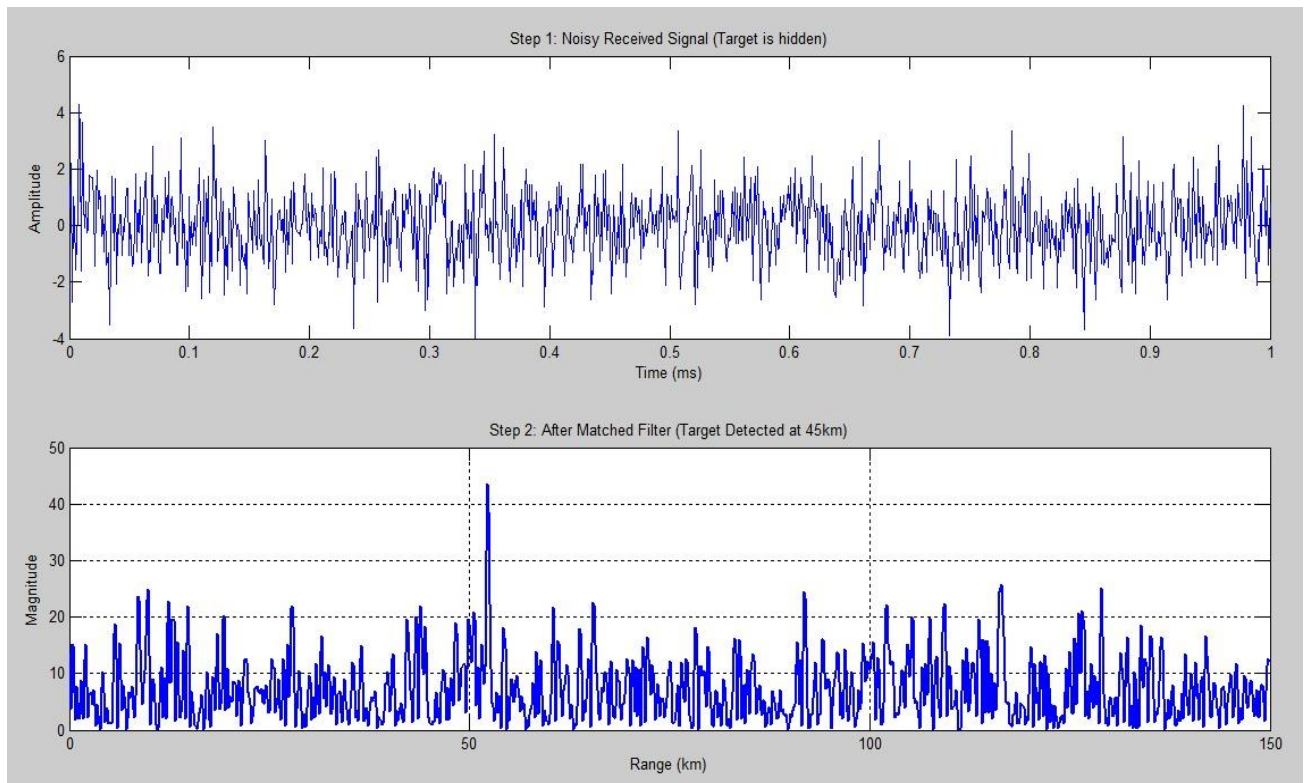
range_axis = (t_total * c) / 2000; % Convert time to Range in km

plot(range_axis, mf_magnitude, 'LineWidth', 1.5);

title('Step 2: After Matched Filter (Target Detected at 45km)');

xlabel('Range (km)'); ylabel('Magnitude'); grid on;
```

Simulation Results:



Applications:

1. Military Surveillance & Defense
2. Air Traffic Control
3. Space Applications
4. Remote Sensing & Mapping

Results:

this project confirms that Matched Filtering is an effective technique for detecting radar signals, even when the reflection is completely buried under noise.

Conclusion:

This project demonstrated how matched filtering can be used for radar signal detection in MATLAB. The matched filter improves the signal-to-noise ratio and enables reliable detection of radar echoes.